

Marco Otero

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Target: Hardware Design Verification Engineering Co-op/Intern

EDUCATION

Georgetown University

Bachelor of Science in Computer Science, First Honors, 3.96 GPA

Washington, DC

Expected May 2028

- **Computer Organization & Architecture, Spring 2026:** logic gates, combinational/sequential logic, FSM, instruction set architecture, assembly programming
- **Circuit Theory, Summer 2026:** op-amp circuit design, RC/RL/RLC transient and steady-state analysis, phasors, power calculations, CAD-based circuit simulation
- **Systems Programming, in progress this Fall 2026:** UNIX system calls, processes/signals, concurrency, shell scripting, memory hierarchy and cache locality, profiling and benchmarking

SKILLS

- Programming: Python, C++, C, Java, Rust, Assembly, Bash, PyTorch
- Hardware/Verification: Verilog, RTL design, gate-level netlists, hardware description languages, digital logic design, ASIC verification benchmark development, fault simulation, ATPG concepts, Yosys synthesis
- Lab Instrumentation: Oscilloscope, function generator, digital multimeter, breadboard prototyping and fault diagnosis
- Tools & Platforms: Git, GitHub, Linux, Raspberry Pi OS, CMake, CLion, Vim

WORK EXPERIENCE

Dr. Radha Venkatagiri's Lab, Computer Science Dept., Georgetown University

Undergraduate Research Assistant

Washington, DC

Jun 2025 - Present

- Wrote test plans and synthesized 16 industry circuits from OpenTitan, covering 345,674 stuck-at faults to achieve 99.2 to 100% fault coverage baseline on each
- Identified 690 Hard-To-Detect faults, 0.2% of all faults, across 6 cryptographic primitives and validated the result with a full regression pass, re-simulating every fault against every existing test vector to confirm zero false positives
- Diagnosed a SystemVerilog parsing failure in Yosys blocking 4 of 16 synthesis targets, then switched the frontend to restore them

Department of Computer Science, Georgetown University

Teaching Assistant, Data Structures

Washington, DC

Aug 2026 - Present

- Graded 5 projects per semester for a 30+ student course, flagging grading and rubric discrepancies to the instructor before they reached students
- Taught data structures and algorithms concepts through weekly office hours and email discussion

PROJECT EXPERIENCE

Two-Channel Powered Audio Docking Station

Capstone Project

May 2026 - Jul 2026

- Built a bass-channel active filter, verifying cutoff frequencies within about 4% of calculated values across a 10 Hz–20 kHz sweep on a Digilent Analog Discovery 2 oscilloscope and function generator
- Diagnosed an LM741 stage's finite output impedance limiting it to 25 mA against a 662 mA peak demand, then added an LM386 power amplifier stage with a cross-rail coupling capacitor to drive the 8Ω speaker load

Cloud Photo Backup System

Infrastructure Engineer

Aug 2026 - Sep 2026

- Designed and deployed a self-hosted photo backup system with a private server on a Raspberry Pi 5, replacing reliance on commercial cloud storage
- Configured Tailscale across multiple devices for secure remote access without port forwarding or public internet exposure

PC Build

PC Technician

Aug 2024 - Present

- Designed and constructed a fully functional custom PC, researching and selecting every component, including the CPU, GPU, case, and cooling, for compatibility, performance, and cost efficiency
- Solved airflow problems by placing intake and outtake fans to improve cooling and performance

LEADERSHIP EXPERIENCE

Office of Campus Ministry, Georgetown University

Ecumenical Student Lead Member

Washington, DC

Apr 2025 - Present

- Coordinated between chaplaincies to organize recurring worship services and community events

First Fellows

Co-Leader, Academic Success & Mentorship

Washington, DC

Jul 2026 - Present

- Designed surveys and events for a peer mentorship program, delivering UX and content recommendations for its website